

# Limited-Memory Renewal Contracts: Exact Pooling and Selected-Book Design

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**Abstract.** A renewal provider must choose a limited command alphabet and allocate an accepted obligation across heterogeneous histories. We develop finite-catalog joint design with opening charges, expectation caps, and realization ceilings. Terminal-first pooling retains individual caps and service costs while assigning an exact correction to the last eligible selected command. Common catalog eligibility therefore permits exact polynomial optimization with a variable symbol budget. Because unused catalog insertions can split eligibility classes, we also develop a mandatory-selected-prefix price recurrence and a certified completion search. This search turns the inherited fixed-budget enumeration regime into a prunable original-budget algorithm without an outer search over all eligibility-class means. We characterize catalog refinement, exact screening, signature-cell stability, and robust design under uncertain ceilings. Explicit dyadic counts support the complementary group-mean accuracy scheme. Independent checks certify original-space policies and complete search covers. Synthetic experiments include strict interior prefixes with 1,024 histories, matched class counts, refinement paths, and direct mixed-integer diagnostics. All six previously difficult instances receive zero-width rational certificates; all interrupted requests are retained. A stochastic capacity-reservation model derives the same correction from expected served demand.

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# 1 Introduction

A renewal contract may promise service at many public histories while its terminal interface implements only a small number of commands. The provider must choose the commands, allocate the promised service across histories, and pay for any intermediate service that cannot be delivered by the terminal command. Individual expectation caps and realization ceilings make these choices interdependent. A command useful at one history may be inadmissible at another; opening a command changes the common interface rather than a single local allocation. The design problem is therefore joint even when allocating service for a fixed interface is easy.

We study this problem on a finite candidate catalog with a fixed installed-symbol budget and nonnegative opening charges. Our principal structural result is terminal-first pooling. Histories admitting the same catalog prefix can be optimized together without averaging their caps or costs. The exact service-cost correction depends on the selected book only through its last eligible level. An ordered path can consequently charge the correction once and jointly optimize the book and the individual targets. When all histories have common catalog eligibility, this yields an exact polynomial algorithm with variable history count, catalog size, and symbol budget. Common eligibility does not require that every catalog level be eligible.

Catalog eligibility is nevertheless not an intrinsic dimension. Adding an unused candidate can split a class without changing the optimum. We address that dependence in two ways. First, we characterize refinement and threshold stability, provide certified deletion rules, and solve threshold-robust design by its lower-ceiling representation. Second, we develop a selected-prefix search. Its state commits to installed symbols rather than a partition of all histories. A mandatory-prefix extension of the two-sided price graph supplies original-budget bounds, and a free-completion bound permits simpler certificates. The resulting binary cover is exact for every fixed symbol budget, regardless of the number of catalog eligibility classes. Fixed-budget polynomial enumeration was already established in the antecedent complexity proposition; the new contribution is the constrained price recurrence and independently checkable pruning, not that elementary enumeration implication. This is a complementary polynomial regime, not a claim of fixed-parameter tractability when the symbol budget itself grows.

The algorithms retain their original-space interpretation at every interruption. A lower policy satisfies the exact aggregate promise, each individual cap, and every supportwise ceiling. Upper certificates combine scalar support prices, Bellman inequalities, and complete decision-tree coverage. Independent verification does not trust an optimizer status or substitute a larger alphabet. We also retain the group-mean box method as a distinct accuracy-based route and give its explicit dyadic cover bound, including initial widths and rounding.

Our experiments separate arithmetic scale from eligibility geometry. Strict interior prefixes, heterogeneous thresholds within a gap, near-boundary perturbations, and successive catalog insertions test the structural claims. We revisit every distinct difficult instance from the preceding study without replacing its inputs or historical failures. Matched-class experiments vary the proposed structural parameter on otherwise fixed primitives. Direct mixed-integer formulations expose solver times, nodes, numerical gaps, and envelope allowances alongside exact rational certificates. All inputs are declared synthetic. A capacity-reservation model derives the same optimization structure from expected served demand rather than asserting empirical

calibration.

The use of capped resource allocation and branch-and-bound is classical. We distinguish these ingredients from the exact last-eligible-level correction, the mandatory-prefix price decomposition, and the resulting original-budget certificate. Existing work on separable resource allocation supplies the inner multiplier machinery (Patriksson and Strömberg, 2015; Schoot Uiterkamp et al., 2022); branch-and-bound supplies the general partition-and-bound principle (Lawler and Wood, 1966). The substantive question here is which exact bounds and primal reconstructions remain valid when command selection, safety eligibility, heterogeneous service costs, and the promise equality are optimized together.

**Relation to quantization and constrained paths.** Ordered interval optimization and scalar quantization already have dynamic-programming and matrix-searching formulations (Wu, 1991; Nielsen and Nock, 2014). Random splitting between grid points preserves a prescribed mean in dual quantization (Pagès and Wilbertz, 2012). Here an adjacent lottery is the canonical local implementation, not the new computational claim. A shared charged catalog, individual realization ceilings, heterogeneous intermediate costs, and an aggregate contractual equality must still be optimized jointly. Our correction retains those individual restrictions while localizing the effect of pooled service at one selected level.

Lagrangian pricing of constrained paths is also established (Handler and Zang, 1980). We do not infer primal exactness from a price optimum or transfer a generic constrained-path hardness result to this restricted model without a reduction. Instead, the mandatory-prefix graph proves an exact support formula for every partial book, and the binary cover supplies the missing primal completeness. Continuous design, Monge acceleration, guarded aggregation, and harmonic coarsening remain complementary results under their original assumptions; the companion’s reading map preserves their complete derivations rather than recasting them as new ingredients of the selected-prefix algorithm.

## 2 Model and Canonical Allocation

There are  $k$  public histories, indexed by  $j = 1, \dots, k$ . History  $j$  is observed with probability  $\pi_j > 0$ , with  $\sum_j \pi_j = 1$ . It has a continuation cap  $b_j \in [0, 1]$  and a convex nondecreasing intermediate-service cost  $h_j$ , with  $h_j(0) = 0$ . The common terminal reward  $f$  is strictly increasing and concave on  $[0, 1]$ . Caps may repeat; write  $\bar{b} = \sum_j \pi_j b_j$ . An encoder selects a terminal command from one installed alphabet, and only its symbol reaches the common terminal gateway. The gateway cannot recover the observed branch, an inherited intermediate tier, or an uncharged shared random seed. Thus  $m$  distinct commands require  $\lceil \log_2 m \rceil$  writable bits; read-only program constants and public-state storage are separate resources. The full institutional mechanism and saturated benchmarks remain in the preserved antecedent article and mathematical archive (the companion gives the reading map).

The provider selects a pre-draw tier  $y_j$  and a lottery  $C_j$  on the installed book, with operating payoff  $\mathbb{E}f(C_j) - h_j(y_j)$ . The accepted root obligation is  $B \in [0, \bar{b}]$ . Expected participation restricts the mean obligation, while a realization ceiling restricts every positive-probability draw. With a common risk tolerance

$\delta \geq 0$ ,

$$t_j = y_j + \mathbb{E}C_j \leq b_j, \quad y_j + C_j \leq b_j + \delta \text{ almost surely}, \quad \sum_j \pi_j t_j = B. \quad (1)$$

A book  $c$  has at most  $m$  distinct levels in  $[0, 1]$  and incurs  $\sum_{z \in c} \rho(z)$ , once per selected level. Let  $\mathcal{V}_m^\delta(B; \rho)$  denote maximum operating payoff less this charge. A finite catalog restriction is denoted  $\mathcal{V}_{m, \mathcal{A}}^\delta(B; \rho)$ . The expectation cap is not replaced by the weaker realization ceiling. We write  $V_m^P(B)$  and  $V_m^E(B)$  for the same zero-charge objective under, respectively, pathwise participation and expectation-only participation, with their encoder restrictions unchanged. Taking  $\delta \geq 1$  makes the realization restriction redundant at the canonical optimum below. Taking  $\delta = 0$  gives the full pathwise value, even though that institution initially permits draw-specific intermediate service. Thus these are endpoints of one design problem, not three unrelated extensions.

## 2.1 Eligibility prefixes replace saturated targets

For a strictly ordered book  $c = (c_1, \dots, c_s)$  and branch  $j$ , define

$$c^{j, \delta} = \{z \in c : z \leq b_j + \delta\}, \quad d_j = \max c^{j, \delta},$$

and let  $F_{j, c}^\delta$  be the piecewise-affine interpolant of  $f$  on this prefix, including the singleton interpretation. The prefix depends on the cap and risk ceiling, not on the endogenous target.

**Theorem 2.1** (Canonical allocation under a realization ceiling). *Suppose  $f$  is strictly increasing and concave and  $h_j$  is convex, nondecreasing, and zero at zero. A fixed book is feasible exactly when  $c_1 \leq \min_j b_j$  and  $c_1 \leq B \leq \bar{b}$ . Its optimal branch response at target  $c_1 \leq t \leq b_j$  is*

$$v_{j, c}^\delta(t) = F_{j, c}^\delta(\min\{t, d_j\}) - h_j((t - d_j)_+). \quad (2)$$

*It is attained by  $y = (t - d_j)_+$  and an adjacent lottery of mean  $\min(t, d_j)$  in the eligible prefix. Consequently*

$$\mathcal{V}_m^\delta(B; \rho) = \max_{\substack{|c| \leq m, \ c_1 \leq t_j \leq b_j \\ \sum_j \pi_j t_j = B}} \left\{ \sum_j \pi_j v_{j, c}^\delta(t_j) - \sum_{z \in c} \rho(z) \right\}. \quad (3)$$

*The fixed-book inner problem is concave. With zero charges,  $\mathcal{V}_m^0(B; 0) = V_m^P(B)$  and  $\mathcal{V}_m^\delta(B; 0) = V_m^E(B)$  for  $\delta \geq 1$ , at every feasible  $B$ .*

*Proof.* Every used terminal level satisfies  $C \leq b_j + \delta$  because  $y \geq 0$ . Writing  $\mu = \mathbb{E}C$ , the payoff is at most  $F_{j, c}^\delta(\mu) - h_j(t - \mu)$ . Both terms are nondecreasing as  $\mu$  increases on  $[c_1, \min(t, d_j)]$ , so this upper bound is maximized at its right endpoint. If  $t \leq d_j$ , the adjacent lottery has  $y = 0$  and all its support lies below  $b_j + \delta$ . If  $t > d_j$ , it is the singleton  $d_j$  and  $y + d_j = t \leq b_j$ . Thus the upper bound is attained without relaxing either constraint. The response is concave: its nonnegative interpolation slopes decrease, and its tail has the nonpositive slopes of  $-h_j(t - d_j)$ . The feasibility conditions follow by using the lowest level and allocating intermediate service; weighted targets fill the entire interval  $[c_1, \bar{b}]$ .

For pathwise participation with possibly draw-specific  $Y$ , all used  $C$  satisfy  $C \leq b_j$ . At fixed  $t = \mathbb{E}(Y+C)$ , concavity of  $f$  on the discrete support and Jensen's inequality for  $h_j$  give the same upper bound with  $\delta = 0$  and  $\mathbb{E}Y = t - \mu$ . The canonical policy attains it and is pathwise feasible. Fresh decoder randomization can first be replaced by its conditional mean for each symbol: this increases terminal reward and preserves all the ceilings under the stipulated absence of an uncharged shared seed. Under expected participation the same mean argument has no eligibility truncation;  $\delta \geq 1$  includes every level in  $[0, 1]$ . These observations establish the two endpoint identities and both directions of (3).  $\square$

For every exact-arithmetic and bit-complexity statement we assume rationally encoded probabilities, caps, ceilings, catalog levels, promise, and nonnegative charges, with

$$f(x) = rx - \frac{q}{2}x^2, \quad r > 0, \quad 0 \leq q \leq r, \quad h_j(y) = \frac{\gamma_j}{2}y^2, \quad \gamma_j \geq 0. \quad (4)$$

The structural identities apply more generally under their stated concavity assumptions; exact bit complexity does not refer to arbitrary real inputs. For these quadratic primitives, chord  $[u, v]$  has slope  $r - q(u + v)/2$ . On branch  $j$  its target capacity is

$$[\min\{b_j, v\} - u]_+ \mathbf{1}\{v \leq b_j + \delta\}.$$

Starting at  $t_j = c_1$ , fill these positive slopes in their common decreasing order, followed by zero-cost intermediate capacity and the quadratic tails. A water-level sweep over events  $\gamma_j(b_j - d_j)_+$  fills the remaining targets. It uses  $O(ks + k \log(k + 1))$  rational arithmetic operations, including the certificate

$$\sum_j \pi_j v_{j,c}^\delta(t_j) = \lambda B + \sum_j \pi_j \max_{c_1 \leq t \leq b_j} \{v_{j,c}^\delta(t) - \lambda t\}. \quad (5)$$

The right-hand branch maxima are independently checked at eligible knots, the cap, and the clipped tail stationary point. This multiplier certifies the allocation for that book; it is not by itself a global codebook certificate. The outer problem is solved next.

All canonical arguments continue to hold for branch-specific ceilings  $\tau_j \geq b_j$  in place of  $b_j + \delta$ : the eligibility prefix is then  $c \cap [0, \tau_j]$ . We write  $\mathcal{V}_{m,\mathcal{A}}^\tau(B; \rho)$  for this generalized catalog value. The new budgeted algorithms and numerical formulation use these heterogeneous ceilings. The retained continuous and institutional results use their stated common- $\delta$  specialization.

### 3 Exact Budgeted Compression at Fixed Targets

We first solve alphabet selection without changing the promised branch targets. This is both an independently meaningful design problem and an oracle for joint design. Let  $\mathcal{A} = (a_1 < \dots < a_N)$ , with charges  $\rho_i \geq 0$ , and fix  $t_j \leq b_j$ ,  $\sum_j \pi_j t_j = B$ , and  $a_1 \leq \min_j t_j$ . Write  $\tau_j \geq b_j$  for the realization ceiling; the renewal specialization is  $\tau_j = b_j + \delta$ . In this section  $v_{j,c}$  denotes (2) with this branch-specific ceiling, and  $R(c) = \sum_{a_i \in c} \rho_i$ .

For two candidate levels  $u < v$ , define the target-crossing edge and stopping reward

$$A_t(u, v) = \sum_{j: u \leq t_j < v} \pi_j \begin{cases} f(u) + \frac{t_j - u}{v - u} \{f(v) - f(u)\}, & v \leq \tau_j, \\ f(u) - h_j(t_j - u), & v > \tau_j, \end{cases} \quad (6)$$

$$Z_t(u) = \sum_{j: t_j \geq u} \pi_j \{f(u) - h_j(t_j - u)\}. \quad (7)$$

The eligibility test is on  $v$ , not on its mean with  $u$ . If it fails, a branch uses the lower level and a fixed tier  $t_j - u$ ; averaging an ineligible realization into a feasible mean would not implement the contract. Unlike the priced recurrence, (6) has no price-admissibility restriction on an arc.

**Theorem 3.1** (Conditional memory–charge frontier). *For any feasible target vector  $t$ , define*

$$F_\ell(t) = \max_{\substack{c \subseteq \mathcal{A}, |c|=\ell \\ c_1 \leq \min_j t_j}} \left\{ \sum_j \pi_j v_{j,c}(t_j) - R(c) \right\}, \quad P_s(t) = \max_{1 \leq \ell \leq s} F_\ell(t).$$

*Infeasible exact cardinalities have value  $-\infty$ . Set  $H_1(i) = Z_t(a_i)$  and*

$$H_\ell(i) = \max_{v > i} \{A_t(a_i, a_v) - \rho_v + H_{\ell-1}(v)\}. \quad (8)$$

*Then  $F_\ell(t) = \max_{a_i \leq \min_j t_j} \{H_\ell(i) - \rho_i\}$ . All frontiers through budget  $M \leq N$  and their feasible policies are computable in  $O((k + M)N^2)$  arithmetic operations and  $O(N^2 + MN)$  storage, given constant-time evaluations of  $f$  and  $h_j$ . Rational quadratic data give polynomial bit complexity. Every policy uses one fixed book, preserves the supplied targets exactly, commits intermediate service before its lottery, and respects every realization ceiling.*

*Proof.* Take a feasible ordered book. Every target belongs to exactly one interval  $[u, v)$  between consecutive levels, or to the final interval  $[u, \infty)$ . In the former case  $u \leq t_j \leq b_j \leq \tau_j$ , so the lower endpoint is eligible. If  $v$  is eligible, the canonical response is its chord at  $t_j$ . If  $v$  is ineligible, every later level is ineligible, and the highest eligible book level is  $u$ ; the response is the second line of (6). A final-interval branch has the response in (7). These intervals partition the branches, including targets equal to a selected level. The book's operating value is therefore the sum of its edge and stopping rewards.

The recurrence enumerates ordered paths with exactly  $\ell$  vertices. It pays every charge once, with the first charge outside the recurrence. Conversely, each such path starting no higher than the smallest target defines a feasible book, and the preceding partition computes its exact value. Maximization proves both directions of the claimed equality. Taking the best exact cardinality gives an at-most budget and does not assume that an unused additional symbol should be purchased. There are  $O(N^2)$  edges, each computed by scanning the branches, and  $O(MN^2)$  transitions. Each rational path value is a sum of polynomially many input-derived rational terms, so ordinary rational arithmetic and comparison have polynomial bit complexity.  $\square$

**Corollary 3.2** (Global catalog design at saturation). *If  $B = \bar{b}$ , then  $P_s(b) = \mathcal{V}_{s,\mathcal{A}}^\tau(\bar{b}; \rho)$  for every  $s$ . Hence joint catalog design at a saturated promise, including arbitrary nonnegative opening charges and*

branch-specific realization ceilings, has the complexity in Theorem 3.1, with both  $k$  and  $s$  variable.

*Proof.* Positive weights and  $t_j \leq b_j$  imply  $\sum_j \pi_j t_j = \bar{b}$  only when every  $t_j = b_j$ . There is no remaining target optimization. Apply the theorem.  $\square$

This corollary is a same-budget global result, not a priced relaxation. At an interior promise, the theorem optimizes the entire alphabet conditional on  $t$ ; it does not certify that the selected target vector is jointly optimal. Section A resolves that target choice at fixed branch count. The faster saturated Monge construction in Appendix E concerns its stronger structural assumptions and is retained rather than replaced.

### 3.1 Compression, opening costs, and symbol prices

Take any candidate pool  $U$ , including a union of supported books. Applying the theorem with  $\mathcal{A} = U$  gives the optimal sub-book at every allowed cardinality, not just a sequence of greedy deletions. Applying it on the whole catalog can only improve the conditional value. No nestedness of optimal books is required. In particular, comparing budgets  $m, m+1, \dots, |U|$  gives the complete conditional memory–value frontier and identifies the value purchased by each additional symbol.

If an additional uniform certification fee  $\kappa \geq 0$  is imposed, the exact conditional value is

$$\max_{1 \leq \ell \leq M} \{F_\ell(t) - \kappa \ell\}. \quad (9)$$

The upper envelope has finitely many breakpoints, obtained from the computed exact-cardinality values. Chosen cardinalities may be tied, but a largest optimal cardinality is nonincreasing in  $\kappa$ . Indeed, optimality at  $\kappa_1 < \kappa_2$  and cardinalities  $\ell_1, \ell_2$  gives  $(\kappa_2 - \kappa_1)(\ell_1 - \ell_2) \geq 0$ . This statement concerns cardinality, not nesting of the selected levels. General nonuniform charges are already included in the recurrence and do not require the uniform-fee interpretation.

The structural advantage over a raw union is that charges enter the optimization itself. A costly level may be discarded even when capacity remains. Nevertheless, an optimal conditional compression can be inferior to a policy with different branch targets. Our returned operational policy therefore compares valid candidates at the *same* budget and keeps the best, including a supplied incumbent. This safeguard never converts a heuristic target portfolio into a global optimality theorem.

## 4 Exact Pooling by Eligibility

Define the catalog signature  $K_j(\mathcal{A}) = |\mathcal{A} \cap [0, \tau_j]|$ . Two histories have the same *eligibility class* when their signatures coincide. This partition depends on the candidate catalog, not only on the primitive contracts. Let  $G_1, \dots, G_E$  be these classes, and write

$$W_g = \sum_{j \in G_g} \pi_j, \quad w_{j|g} = \pi_j / W_g, \quad \bar{b}_g = \sum_{j \in G_g} w_{j|g} b_j.$$

This partition does not require identical caps or costs. In particular, an arbitrarily large collection with common catalog eligibility has  $E = 1$ , even if every cap and every curvature differs. We retain each original branch when evaluating a class; we do not replace it by an artificial branch with an average cost. The group mean  $s_g$  satisfies  $\sum_g W_g s_g = B$ .

For  $0 \leq s \leq \bar{b}_g$ , define the capped equalization

$$\theta_j^g(s) = \min\{b_j, z_g(s)\}, \quad \sum_{j \in G_g} w_{j|g} \min\{b_j, z_g(s)\} = s. \quad (10)$$

At  $s = \bar{b}_g$ , use  $\theta_j^g(s) = b_j$ . Although the water level is then not unique, the allocation is unique. Each coordinate is nondecreasing in  $s$ . For a book  $c$  with a feasible anchor, let  $d_g$  be its last level eligible for this class and let  $F_{g,c}$  be its common reward interpolant. Define

$$D_g(d) = \sum_{j \in G_g} w_{j|g} \min\{b_j, d\}, \quad (11)$$

$$C_g(q; d) = \min \left\{ \sum_{j \in G_g} w_{j|g} h_j(y_j) : 0 \leq y_j \leq (b_j - d)_+, \sum_{j \in G_g} w_{j|g} y_j = q \right\}. \quad (12)$$

The second problem is ordinary separable resource allocation, not a new water-filling algorithm. Breakpoint and multiplier methods for such problems are established; see Patriksson and Strömberg (2015) and Schoot Uiterkamp et al. (2022). What matters here is that its only book-dependent argument is the last eligible level.

**Theorem 4.1** (Terminal-first pooling). *For continuous convex nondecreasing costs that vanish at zero, the maximum normalized payoff of class  $g$  at mean  $s \in [c_1, \bar{b}_g]$  equals*

$$V_{g,c}(s) = \begin{cases} \sum_{j \in G_g} w_{j|g} F_{g,c}(\theta_j^g(s)), & s \leq D_g(d_g), \\ \sum_{j \in G_g} w_{j|g} F_{g,c}(\min\{b_j, d_g\}) - C_g(s - D_g(d_g); d_g), & s \geq D_g(d_g). \end{cases} \quad (13)$$

Consequently,

$$W_g V_{g,c}(s) = \sum_{j \in G_g} \pi_j v_{j,c}^\tau(\theta_j^g(s)) + \Delta_g(d_g; s), \quad (14)$$

$$\Delta_g(d; s) = \sum_{j \in G_g} \pi_j h_j((\theta_j^g(s) - d)_+) - W_g C_g((s - D_g(d))_+; d) \geq 0. \quad (15)$$

Below  $D_g(d_g)$  an optimal allocation is  $t_j = \theta_j^g(s)$ ; above it an optimal allocation is  $t_j = \min\{b_j, d_g\} + y_j$ , where  $y$  solves (12). These policies satisfy the individual caps and realization ceilings, not merely their group averages.

*Proof.* Use the canonical response in Theorem 2.1. Suppose a feasible allocation gives positive intermediate service to one member while another has unused terminal capacity below  $\min\{b_j, d_g\}$ . If member  $i$  has positive service and member  $j$  has spare terminal capacity, choose  $\delta > 0$  small enough that both

capacity inequalities remain valid and set  $t'_i = t_i - \delta$ ,  $t'_j = t_j + (\pi_i/\pi_j)\delta$ . The conservation equation  $\pi_i(t'_i - t_i) + \pi_j(t'_j - t_j) = 0$  keeps the class mean unchanged. Intermediate cost cannot increase, and the strictly increasing common interpolant raises terminal reward. Thus no optimum mixes positive service with unfinished terminal capacity.

If  $s \leq D_g(d_g)$ , an optimum therefore uses no service. Maximizing the same concave interpolant over capped targets gives (10). To verify this without assuming differentiability, take a supporting slope at the common unsaturated level. On an unsaturated member its supporting-line inequality bounds any change in reward. On a capped member, decreasing its target loses at least that slope per unit, by concavity. The weighted linear terms sum to zero. At the endpoint where all terminal capacities are full, the assertion follows directly by monotonicity. This proves the first case of (13).

If  $s \geq D_g(d_g)$ , every terminal capacity is full and only service remains to be allocated. Its capacities and total are precisely those in (12), proving the second case. At  $s = D_g(d_g)$ , both formulas equal  $\sum_j w_{j|g} F_{g,c}(\min\{b_j, d_g\})$ , because  $C_g(0; d_g) = 0$ ; continuity of the finite capped convex allocation value gives continuity at this boundary. If  $s > D_g(d_g)$  and the equalization level were at most  $d_g$ , its weighted mean would be at most  $D_g(d_g)$ , a contradiction. Hence  $\theta_j^s(s) \geq \min\{b_j, d_g\}$ , so the equalized allocation has the same terminal rewards as the optimal allocation. Their difference is exactly the reduction in service cost in (15). Below  $D_g(d_g)$  both cost terms are zero. Feasibility follows from the original cap constraints and the canonical adjacent lottery or singleton, including its supportwise ceiling.  $\square$

#### 4.1 Joint book selection at fixed group means

Let  $K_g$  be the number of catalog levels eligible for class  $g$ . For supplied group means, form the individual targets  $t_j = \theta_j^g(s_g)$  and the target-crossing edges and tails in Section 3. Add

$$\sum_{g:i \leq K_g < v} \Delta_g(a_i; s_g) \quad (16)$$

to the edge from catalog index  $i$  to  $v$ , and add  $\sum_{g:i \leq K_g} \Delta_g(a_i; s_g)$  to a path ending at  $i$ . Indices in this formula are one-based. The correction is paid at the unique edge that leaves a class's eligible prefix, or at the terminal node if no such edge occurs. All original level-opening charges remain in the path objective.

**Theorem 4.2** (Exact eligibility-mean frontier). *The corrected path recurrence returns the globally best book and individual targets for supplied feasible group means, at every budget up to  $m$ . For rationally encoded nonnegative opening charges and rational quadratic primitives  $f(x) = rx - qx^2/2$  and  $h_j(y) = \gamma_j y^2/2$ , including zero service curvatures, one call takes*

$$O((k+m)N^2 + kN \log(k+1)) \quad (17)$$

*rational arithmetic operations, with polynomial bit complexity in the instance and supplied means. When  $E = 1$ , set  $s_1 = B$ : a single call solves the original joint finite-catalog problem globally, for variable  $k, N, m$ , nonnegative opening charges, and heterogeneous caps and curvatures. Optimization identities also permit*

*real charges; the bit-complexity assertion requires the rational encoding just specified. It does not enumerate books or expand the command budget.*

*Proof.* For any book, the uncorrected path sums its responses at the equalized targets. Every class has exactly one last eligible selected symbol. Formula (16) adds its correction exactly once, so Theorem 4.1 makes the path value equal to that book's optimal value at the group means. Conversely every path defines a legal book and the theorem constructs its optimal individual allocation. Maximizing paths therefore proves both inequalities needed for exactness.

For each possible last level, solve (12) by first filling zero-cost capacity and then sorting the positive breakpoints  $\gamma_j(b_j - d)_+$ . The remaining water level solves a linear equation between consecutive breakpoints. This requires  $O(|G_g| \log(|G_g| + 1))$  arithmetic operations per class and level. Compute equalized targets once, construct the original  $O(kN^2)$  crossing terms, and add all class corrections in  $O(EN^2)$  operations. Since  $E \leq k$ , the path recurrence and these costs give (17). All sums, comparisons, divisions, and reconstructed probabilities are rational with polynomial encoding length. For  $E = 1$  the root promise already fixes the only group mean. Thus there is no outer target search, including at interior promises.  $\square$

This result differs from aggregation of identical response types in Proposition EC.2.1. It retains arbitrary within-class heterogeneity and evaluates its exact service frontier. It also differs from harmonic coarsening: no cap guard or dispersion allowance is needed. The earlier harmonic and minimum-curvature results remain useful simpler upper models under their stated assumptions and remain available in the preserved mathematical archive.

## 5 An Exact Two-Sided Price Oracle

After any exact type reduction, retain the index  $j$  for a branch and its weight  $\pi_j$ . A target box  $X = [\ell, u]$  satisfies  $0 \leq \ell_j \leq u_j \leq b_j$  and intersects  $\sum_j \pi_j t_j = B$ . Only anchors in

$$I_X = \left\{ i : a_i \leq \min_j u_j, \quad \sum_j \pi_j \max\{\ell_j, a_i\} \leq B \right\} \quad (18)$$

can meet the equality inside this box. Because  $B \leq \sum_j \pi_j u_j$ , these conditions are also sufficient for a fixed book's box feasibility. They depend on its lowest level, not on its remaining levels.

Let  $g(x) = f(x) - \lambda x$  at a fixed real price  $\lambda$ . For a book whose first index lies in  $I_X$ , relax only the aggregate equality and define

$$S_X(\lambda) = \max_{\substack{\emptyset \neq c \subseteq \mathcal{A}, |c| \leq m \\ c_1 = a_i, i \in I_X}} \left\{ \sum_j \pi_j \max_{\max\{\ell_j, c_1\} \leq t \leq u_j} [v_{j,c}(t) - \lambda t] - R(c) \right\}, \quad (19)$$

$$U_X(\lambda) = \lambda B + S_X(\lambda). \quad (20)$$

Every  $U_X(\lambda)$  bounds the best original-budget joint design in  $X$ . No zero-duality-gap assumption is made.

For a general cost define the clipped scalar support

$$\Phi_j(s, t; \lambda) = \max_{s \leq y \leq t} \{-h_j(y) - \lambda y\}.$$

For quadratic costs it is obtained by clipping  $-\lambda/\gamma_j$  to  $[s, t]$  when  $\gamma_j > 0$ ; for zero curvature choose the endpoint with the better linear value. Write  $\psi_j(z) = \Phi_j(\max\{0, \ell_j - z\}, u_j - z; \lambda)$  whenever  $z \leq u_j$ .

## 5.1 The increasing and decreasing sides

For  $x < z$  define an upper-crossing edge and its terminal value by

$$A^+(x, z) = \sum_{j: x \leq u_j < z} \pi_j \begin{cases} g(x) + \frac{u_j - x}{z - x} [g(z) - g(x)], & z \leq \tau_j, \\ g(x) + \psi_j(x), & z > \tau_j, \end{cases} \quad (21)$$

$$T^+(x) = \sum_{j: u_j \geq x} \pi_j [g(x) + \psi_j(x)]. \quad (22)$$

When  $\lambda > 0$  also define a lower-crossing edge, a lower terminal value, and a peak value:

$$A^-(x, z) = \sum_{j: x \leq \ell_j < z} \pi_j \begin{cases} g(x) + \frac{\ell_j - x}{z - x} [g(z) - g(x)], & z \leq \tau_j, \\ g(x) - h_j(\ell_j - x) - \lambda(\ell_j - x), & z > \tau_j, \end{cases} \quad (23)$$

$$T^-(x) = \sum_{j: \ell_j \geq x} \pi_j [g(x) - h_j(\ell_j - x) - \lambda(\ell_j - x)], \quad (24)$$

$$H(x) = \sum_{j: \ell_j < x \leq u_j} \pi_j g(x). \quad (25)$$

Only increasing-score arcs use  $A^+$ , and only decreasing-score arcs use  $A^-$ . Equal-score arcs may be assigned to either side. Set missing forward states to  $-\infty$ , and compute

$$\alpha_1(i) = \begin{cases} -\rho_i, & i \in I_X, \\ -\infty, & \text{otherwise,} \end{cases} \quad (26)$$

$$\alpha_r(v) = -\rho_v + \max_{\substack{i < v \\ g(a_i) \leq g(a_v)}} [\alpha_{r-1}(i) + A^+(a_i, a_v)], \quad 2 \leq r \leq m.$$

This table pays the charge for its last level. For positive prices a backward table excludes the charge for its first level:

$$\begin{aligned} \beta_1(i) &= T^-(a_i), \\ \beta_s(i) &= \max \left\{ T^-(a_i), \max_{\substack{v > i \\ g(a_v) \leq g(a_i)}} [A^-(a_i, a_v) - \rho_v + \beta_{s-1}(v)] \right\}, \quad 2 \leq s \leq m. \end{aligned} \quad (27)$$

**Theorem 5.1** (Two-sided box-price decomposition). *If  $I_X$  is nonempty, the exact support in (19) is*

$$S_X(\lambda) = \begin{cases} \max_{1 \leq r \leq m, i} \{\alpha_r(i) + T^+(a_i)\}, & \lambda \leq 0, \\ \max_{1 \leq r \leq m, i} \{\alpha_r(i) + H(a_i) + \beta_{m-r+1}(i)\}, & \lambda > 0. \end{cases} \quad (28)$$

*The maximizing book and scalar target supports are recoverable. Computing one price bound takes  $O((d+m)N^2)$  arithmetic operations and  $O(N^2 + mN)$  storage, where  $d$  is the number of branches after type reduction. General costs additionally require  $O(dN^2)$  clipped scalar-oracle calls. Rational quadratic data and a rational price give an exact rational bound in polynomial bit complexity for each node.*

*Proof.* Concavity of  $g$  implies that its values on any ordered book increase to a peak and then decrease; a tie can be placed on either side. For  $\lambda \leq 0$ , strict increase of  $f$  makes all scores strictly increasing, so the peak is the final level. Consider first a positive price and a chosen peak  $p$ .

If  $u_j < p$ , exactly one increasing edge  $[x, z)$  crosses  $u_j$ . If  $z$  is eligible, the interpolant of  $g$  is nondecreasing through this edge and is maximized on the permitted target interval at  $u_j$ . No later segment below  $u_j$  exists. If  $z$  is ineligible, every later book level is ineligible, so  $x$  is the last usable level and clipped shortfall maximization gives  $g(x) + \psi_j(x)$ . This proves the upper-crossing formula. Its clipping is essential when  $\ell_j > x$ .

If  $\ell_j < p \leq u_j$ , the peak is eligible because  $p \leq u_j \leq b_j \leq \tau_j$ . It attains  $g(p)$  at zero shortfall. All eligible chord values are at most  $g(p)$ , and positive-price shortfall cannot improve them. This proves the middle term.

If  $\ell_j \geq p$ , the interpolant on the decreasing side is nonincreasing. Each shortfall tail is also nonincreasing because  $h_j$  is nondecreasing and  $\lambda > 0$ . The smallest permissible target,  $\ell_j$ , therefore maximizes the branch's priced payoff. Its crossing edge gives (23); when the next level is ineligible it uses the singleton plus the required shortfall. Targets beyond the last level give (24). Equality at the peak belongs to this third group, preventing double counting.

These three groups partition the branches. Upper crossings before the peak, the peak, and lower crossings after the peak thus decompose the exact priced value of every book, not merely a dominant subset of books. Forward states enumerate all increasing prefixes starting at a feasible anchor. Backward states enumerate all decreasing suffixes, with an optional stop. Gluing a prefix of  $r$  levels to a suffix of at most  $m - r + 1$  levels counts the shared peak once, counts its charge once, and uses at most  $m$  levels. Every feasible ordered book has such a representation, and every represented path is a feasible book for the box. This proves the positive-price formula.

For nonpositive prices, the best point on an eligible increasing chord up to the cap is the cap. Where the next level is ineligible, the last singleton plus clipped shortfall gives the scalar optimum; the same argument gives the final tail. Hence (21) and (22) partition all branches, proving the first case.

The edge construction scans at most  $d$  branches for each ordered pair. Each table has  $mN$  states and at most  $N$  transitions per state. Backpointers reconstruct the book, and scalar maximization reconstructs its supports. For quadratic rational inputs each edge uses a polynomial number of rational operations; each path sums polynomially many terms. Their numerator and denominator lengths remain polynomial in the input,

price, and box-endpoint bit lengths. This is a per-node bit-complexity statement, not a bound independent of search depth.  $\square$

**Why a second side is necessary.** Take equal weights, targets and caps  $(1/5, 1/2, 4/5)$ , unit shortfall curvature, unit ceilings, catalog  $\{1/5, 1/2, 4/5\}$ , zero charges, budget three, and reward  $f(x) = 2x - x^2/2$ . At price  $3/2$ , the complete book has scores  $(2/25, 1/8, 2/25)$ . On the singleton target box its exact joint value is  $169/200$ . Restricting to nondecreasing-score books gives only  $83/100$ , a shortfall of  $3/200$ . Thus the unrestricted deletion argument in Theorem B.1 cannot be reused on a target box. The new recurrence repairs that structural obstruction rather than placing the old one-sided oracle inside a generic partitioning procedure.

## 6 Certified Search in Eligibility Means

For several eligibility classes, the joint search can use their means instead of individual targets. Let  $X = [\ell, u]$  be a group-mean box intersecting  $\sum_g W_g s_g = B$ , with  $0 \leq \ell_g \leq u_g \leq \bar{b}_g$ . A feasible book anchor  $a$  must satisfy

$$a \leq \min_j b_j, \quad a \leq \min_g u_g, \quad \sum_g W_g \max\{\ell_g, a\} \leq B. \quad (29)$$

These conditions are sufficient as well: all individual caps are at least  $a$ , and the group-mean intervals fill every total between their lower and upper sums. Define the artificial individual box

$$l_j = \theta_j^g(\ell_g), \quad v_j = \theta_j^g(u_g), \quad j \in G_g. \quad (30)$$

Its feasible-anchor test is equivalent to (29). The artificial box is only an algebraic input to the existing two-sided oracle; we correct its priced value before treating it as a group-box bound.

### 6.1 The exact service correction at a price

Fix a real price  $\lambda$  and a possible last eligible level  $d$ . If  $u_g < D_g(d)$ , set  $\Xi_g(d; X, \lambda) = 0$ . Otherwise, put

$$q_- = (\ell_g - D_g(d))_+, \quad q_+ = u_g - D_g(d), \quad c_j = (b_j - d)_+,$$

and define

$$T_g(d; X, \lambda) = \max_{\substack{0 \leq y_j \leq c_j \\ q_- \leq \sum_j w_{j|g} y_j \leq q_+}} \sum_{j \in G_g} w_{j|g} \{-h_j(y_j) - \lambda y_j\}, \quad (31)$$

$$J_g(d; X, \lambda) = \sum_{j \in G_g} w_{j|g} \max_{(l_j - d)_+ \leq y \leq (v_j - d)_+} \{-h_j(y) - \lambda y\}, \quad (32)$$

$$\Xi_g(d; X, \lambda) = W_g [T_g(d; X, \lambda) - J_g(d; X, \lambda)]. \quad (33)$$

Every vector admitted by  $J_g$  has mean service in  $[q_-, q_+]$ . Thus  $\Xi_g \geq 0$ . Quadratic  $T_g$  is computed by clipping its unconstrained total to  $[q_-, q_+]$  and solving the capped-cost problem at that total.

**Theorem 6.1** (Exact group-box price decomposition). *For every book with an anchor satisfying (29),*

$$W_g \max_{\max\{\ell_g, c_1\} \leq s \leq u_g} \{V_{g,c}(s) - \lambda s\} = \sum_{j \in G_g} \pi_j \max_{\max\{l_j, c_1\} \leq t \leq v_j} \{v_{j,c}^\tau(t) - \lambda t\} + \Xi_g(d_g; X, \lambda). \quad (34)$$

Apply the two-sided recurrence of Theorem 5.1 to (30). Add  $\sum_{g:i \leq K_g < v} \Xi_g(a_i; X, \lambda)$  to both its rising and falling edges, and add  $\sum_{g:i \leq K_g} \Xi_g(a_i; X, \lambda)$  to both terminal formulas. Leave its peak term and charges unchanged. The resulting support  $S_X^{\text{elig}}(\lambda)$  is exact over all original-budget books and group-box targets. Hence

$$U_X^{\text{elig}}(\lambda) = \lambda B + S_X^{\text{elig}}(\lambda) \quad (35)$$

is a global upper bound for the joint problem in  $X$ , with the per-call arithmetic bound (17) for rational quadratic primitives.

*Proof.* If  $u_g < D_g(d_g)$ , all useful means lie in the terminal-only range. Capped equalization in Theorem 4.1 attains the group optimum at every such mean, and its coordinates lie in the artificial box. Conversely every vector in that box has an allowed group mean and is an original feasible allocation. The two priced suprema are equal and no correction is needed.

Suppose  $u_g \geq D_g(d_g)$ . For  $\lambda \leq 0$ , terminal reward minus price is strictly increasing. Every member's terminal capacity is therefore filled before service is chosen. The true group problem has residual service interval  $[q_-, q_+]$ , whereas the artificial box has the separate service bounds in  $J_g$ . Their terminal reward and price constants coincide; their difference is (33).

For  $\lambda > 0$  and  $\ell_g \leq D_g(d_g)$ , neither problem needs service at an optimum. Reducing service strictly improves the priced objective, and an allowed terminal-only allocation exists. The capped-equalization argument again gives equality of the priced suprema; both  $T_g$  and  $J_g$  are zero. Finally, if  $\lambda > 0$  and  $\ell_g > D_g(d_g)$ , service is unavoidable. Each problem chooses its smallest allowed mean. Their terminal parts coincide and their service difference is again  $T_g - J_g$ . The anchor cannot alter a service interval in these cases because  $c_1 \leq d_g$  and  $c_1 \leq \min_j b_j$ .

This proves (34) for every sign and boundary case, including  $\lambda = 0$ . As in Theorem 4.2, the last eligible level is assigned to exactly one crossing edge or terminal node. The two-sided recurrence represents every book through the unimodality of the common level scores, so the added correction changes its value without restricting its paths. Weak duality gives (35). Computing all service corrections costs  $O(kN \log(k+1))$  operations; the inherited edge and path costs give the stated bound.  $\square$

**Service endpoints.** For  $u_g \geq D_g(d)$ , the equalized coordinates satisfy  $\theta_j^g(u_g) \geq \min\{b_j, d\}$ . At equality this follows directly from the filled terminal capacities, including the case in which every cap is below  $d$  and

the water level is not unique. Hence

$$\sum_{j \in G_g} w_{j|g} (\theta_j^g(u_g) - d)_+ = \sum_{j \in G_g} w_{j|g} (\theta_j^g(u_g) - \min\{b_j, d\}) = u_g - D_g(d).$$

The lower endpoint similarly equals  $(\ell_g - D_g(d))_+$ : it is zero below the threshold and  $\ell_g - D_g(d)$  above it. Therefore every vector in the artificial individual service intervals has aggregate service in  $[q_-, q_+]$ . This explicitly verifies the inclusion used in the group-support correction, including either endpoint at the threshold.

## 6.2 Accuracy and independent certificates

The group response  $V_{g,c}$  is  $L$ -Lipschitz in its mean whenever all individual responses are  $L$ -Lipschitz. Indeed, increasing or decreasing the mean of a feasible original allocation by  $\eta$  can be implemented by bounded transfers in one direction with weighted absolute change  $|\eta|$ . Applying the individual Lipschitz bounds and reversing the comparison proves the claim. For the quadratic model one may take  $L = \max\{r, \max_j \gamma_j\}$ .

**Theorem 6.2** (Eligibility-parameterized joint accuracy). *For  $E > 1$  and  $\varepsilon > 0$ , exact conditional optimization on a repaired net of group means returns an original-feasible policy with loss at most  $\varepsilon$ . A sufficient number of oracle calls is*

$$N \left( 1 + \frac{2L(E-1)}{\varepsilon} \right)^{E-1}. \quad (36)$$

*Alternatively, projected group-box subdivision using (35), with zero included among the tested prices and exact recovery of each returned book, gives a valid rational interval at every interruption. A completed finite subdivision attains any positive tolerance. Choose one dependent coordinate and write  $w_g^0 = W_g(u_g^0 - \ell_g^0)$  for each initial free-coordinate width. Put*

$$\eta_{\text{sub}} = \frac{\varepsilon}{2L(E-1)}, \quad q_g = \begin{cases} 0, & w_g^0 \leq \eta_{\text{sub}}, \\ \lceil \log_2(w_g^0/\eta_{\text{sub}}) \rceil, & w_g^0 > \eta_{\text{sub}}. \end{cases} \quad (37)$$

*Its number of leaves is at most the exact dyadic product*

$$M_{\text{dyad}} = \prod_{g \neq g_0} 2^{q_g} \leq \left( \max \left\{ 1, \frac{4L(E-1)}{\varepsilon} \right\} \right)^{E-1}, \quad (38)$$

*and its number of nodes is at most  $2M_{\text{dyad}} - 1$ . Split only a free coordinate whose width exceeds  $\eta_{\text{sub}}$ ; choosing the widest such coordinate is sufficient. For  $E = 1$ , Theorem 4.2 gives an exact single-call solution instead of either search.*

*Proof.* For each feasible catalog anchor  $a$ , discretize  $E - 1$  weighted group means  $W_g s_g$  in their intervals  $[W_g a, W_g \bar{b}_g]$  with step  $\eta_{\text{net}} = \varepsilon/[2L(E-1)]$ . Set the final mean to the remaining promise, clip it to its bounds, and repair any residual by bounded transfers. These are operations on group means; Theorem 4.2 subsequently allocates the original branches exactly.

Consider an optimal book with anchor  $a$  and means  $s^*$ . Rounding down its free weighted coordinates removes less than  $(E - 1)\eta_{\text{net}}$  total mass. The dependent coordinate and subsequent upward repair restore precisely that mass without leaving the group caps. The weighted absolute distance from  $s^*$  is therefore less than  $2(E - 1)\eta_{\text{net}}$ . The same optimal book remains feasible. The group Lipschitz bound limits its loss to  $\varepsilon$ , and the exact conditional oracle cannot do worse. Each free coordinate has weighted range at most one; enumeration of anchors gives (36).

For subdivision, let  $w(X) = \sum_g W_g(u_g - \ell_g)$ . An optimal zero-price relaxed book has group targets in  $X$ . Its anchor satisfies (29), so bounded transfers can restore the root equality within the same box and book. The loss is at most  $Lw(X)$ . Exact allocation of that returned book at the root promise does at least as well. Therefore a node cannot retain a certification gap greater than  $Lw(X)$ .

Project each box onto the root equality and choose one dependent coordinate. Its weighted width is no larger than the sum of the free weighted widths. Consequently  $w(X)$  is at most twice that sum. If every free width is at most  $\eta_{\text{sub}}$ , the gap is at most  $2L(E - 1)\eta_{\text{sub}} = \varepsilon$ . Projection never enlarges a coordinate interval, and each split of coordinate  $g$  halves its current width. Thus it can be split at most  $q_g$  times on any root-to-leaf path. Every path has depth at most  $\sum_{g \neq g_0} q_g$ , so a binary tree has at most  $2^{\sum q_g} = M_{\text{dyad}}$  leaves and  $2M_{\text{dyad}} - 1$  nodes. For a positive ratio  $x > 1$ ,  $2^{\lceil \log_2 x \rceil} < 2x$ ; for  $x \leq 1$  no split is needed. Using  $w_g^0 \leq 1$  gives the displayed coarse bound. Zero widths contribute a factor one and never require a logarithm. The covering argument and weak duality remain valid before completion.  $\square$

The product, rather than its coarse simplification, records the actual initial geometry. For instance, an interval of width one refined to cutoff  $1/3$  requires four dyadic leaves, not three. A stricter convention  $\eta_{\text{sub}} = \varepsilon/[4L(E - 1)]$  is also valid but must use its own dyadic product (and the corresponding factor-eight coarse bound). Net spacing and subdivision cutoff must not be silently interchanged.

The exponent in this particular scheme is the sufficient catalog-dependent dimension  $E - 1$ , not an intrinsic lower bound on the problem's complexity. Section 7 develops a complementary algorithm whose exponent depends on the installed-symbol budget instead. At fixed  $E$  and bounded primitive scale, this is an additive scheme polynomial in the history count, catalog size, and inverse accuracy. It is not a relative-error approximation assertion, nor a polynomial bound when eligibility complexity varies. The exact common-eligibility result is stronger: it needs no tolerance parameter at all.

For independent checking, a service allocation in (31) is accompanied by one rational multiplier  $\nu$ . Its upper witness is

$$T_g \leq \nu \begin{cases} q_+, & \nu \geq 0, \\ q_-, & \nu < 0 \end{cases} + \sum_{j \in G_g} w_{j|g} \max_{0 \leq y \leq c_j} \{-h_j(y) - (\lambda + \nu)y\}. \quad (39)$$

The checker evaluates each scalar maximum at endpoints and the clipped quadratic stationary point. Equality with the supplied feasible service vector verifies the correction exactly. It then checks both path-table inequalities, every feasible anchor, the complete binary group-box cover, and the original uneliminated policy. It does not import the pooling optimizer or trust a completion flag. This closes the chain from an exact service allocation to an original-space joint-design interval.

## 7 Selected-Prefix Search at the Original Symbol Budget

The preceding algorithm fixes catalog eligibility classes before selecting a book. A complementary approach fixes selected levels and leaves all remaining distinctions endogenous. This distinction matters when a small installed alphabet is selected from a refined catalog. We develop exact upper bounds for this search, including the opening charges, rather than applying a target grid after selecting the book.

### 7.1 A fixed-book prerequisite and endogenous pooling

For a selected book  $c$ , write  $F_c$  for the interpolant on the entire book,  $d_j(c) = \max(c \cap [0, \tau_j])$ , and  $e_j(c) = \min\{b_j, d_j(c)\}$ . Although two histories may have different eligible prefixes, their terminal interpolants agree wherever both are defined: they are restrictions of  $F_c$ . The inherited canonical allocation therefore has the following useful representation. Put  $D(c) = \sum_j \pi_j e_j(c)$ . If  $B \leq D(c)$ , choose  $t_j = \min\{e_j(c), z\}$  with  $\sum_j \pi_j t_j = B$ . If  $B > D(c)$ , choose

$$t_j = e_j(c) + y_j, \quad y \in \arg \min \left\{ \sum_j \pi_j h_j(y_j) : 0 \leq y_j \leq b_j - e_j(c), \sum_j \pi_j y_j = B - D(c) \right\}. \quad (40)$$

The terminal-first exchange and supporting-slope argument in Theorem 4.1, now applied to the common restricted interpolant and effective caps  $e_j(c)$ , prove this representation. It is a restatement and implementation of the fixed-book sweep following Theorem 2.1, not a new claim of novelty for resource allocation. For rational quadratic data it costs  $O(k|c| + k \log(k+1))$  operations. The supporting multiplier in (5) certifies the result.

The number of distinct eligible prefixes *on this book* is at most  $|c|$ , whatever the full-catalog count. This observation does not authorize replacing the whole problem by the winning book's partition before optimization. Different competing books induce different partitions. The following search handles that dependence explicitly.

### 7.2 Mandatory prefixes in the price graph

Order the catalog as  $a_1 < \dots < a_N$ . A binary search state has already accepted a prefix of selected indices  $P$ , rejected the other processed indices, and leaves an undecided suffix  $R$ . Write  $S = P \cup R$  in increasing order. In the local order on  $S$ , its first  $p = |P|$  candidates are mandatory. A completion contains all these candidates and at most  $m$  levels altogether. Its anchor must still satisfy  $c_1 \leq \min_j b_j$  and  $c_1 \leq B$ .

Apply the two-sided oracle in Section 5 on the original target box  $[0, b]$ , making only the following restrictions. If  $p > 0$ , the forward table starts at the first local candidate. An edge from local index  $i$  to  $v$  is admissible only when

$$i \geq p \quad \text{or} \quad v = i + 1. \quad (41)$$

A tail is admissible only at  $i \geq p$ . In particular, a backward stopping value before the final mandatory candidate is infeasible, not zero. The rising and falling score tests and all level charges are unchanged. Denote the resulting priced support by  $S_{P,R}^m(\lambda)$ .

**Theorem 7.1** (Mandatory-prefix price decomposition). *The restricted recurrence computes exactly*

$$S_{P,R}^m(\lambda) = \max_{\substack{P \subseteq c \subseteq S, 1 \leq |c| \leq m \\ c_1 \leq \min_j b_j, c_1 \leq B}} \left\{ \sum_j \pi_j \max_{c_1 \leq t \leq b_j} [v_{j,c}^\tau(t) - \lambda t] - \sum_{x \in c} \rho(x) \right\}. \quad (42)$$

Thus  $U_{P,R}(\lambda) = \lambda B + S_{P,R}^m(\lambda)$  is an upper bound on every completion at the original promise and symbol budget. A call takes  $O((k+m)|S|^2)$  rational operations for rational quadratic data and a rational price. It requires no full-catalog eligibility partition.

*Proof.* A path beginning at the first mandatory candidate cannot skip a subsequent mandatory candidate: an edge before index  $p$  must advance by exactly one. Conversely, every completion containing all mandatory candidates obeys this rule, because these candidates precede every undecided candidate. Stopping is permissible exactly after all mandatory candidates have been visited. These statements also apply when the score peak occurs inside the mandatory prefix: its remaining mandatory levels belong to the decreasing side and their charges must still be paid.

The ownership proof of Theorem 5.1 decomposes the priced response of *each* feasible book into upper crossings, a peak, and lower crossings. Restricting its path family as above therefore preserves equality book by book. Every feasible completion has a represented path, and every represented path is a feasible completion. Taking maxima proves (42); adding  $\lambda B$  proves weak duality without presuming zero duality gap. The restrictions remove edges and stopping options but add no state coordinate. The same edge construction and two tables give the stated operation count. All table entries sum a polynomial number of rational primitive terms.  $\square$

Prices can be generated locally. A maximizing book gives a smallest and largest supported total target at that price. If both exceed  $B$ ,  $U_{P,R}$  has a nonpositive supported slope and the next trial moves upward in price; if both lie below  $B$ , it moves downward. If  $B$  lies between them, bounded transfers among tied maximizers implement the promise and close that node's primal-dual gap. Otherwise a finite number of bisections supplies valid upper bounds even without convergence. For the implemented quadratic primitives,  $[-\max_j \gamma_j - 1, r + 1]$  is a sufficient initial bracket. Ties between books do not invalidate this rule: a supporting slope of any maximizing book is a subgradient of the maximum. Our correctness argument uses only the evaluated bounds, not an assumption that the sampled prices attain their minimum.

### 7.3 Free completions, screening, and finite coverage

Let  $G(S)$  be the gross fixed-book value when all symbols in  $S$  are available without an opening charge. Equation (5) certifies  $G(S)$ . Every completion is feasible in this enlarged command set, so

$$\bar{U}(P, R) = G(S) - \sum_{x \in P} \rho(x) \quad (43)$$

is an additional upper bound. At a state with  $|P| = m$ , use  $S = P$ : optional symbols cannot appear in any completion. A nonempty selected prefix itself gives a feasible lower policy whenever its anchor is feasible.

We retain the smaller of the free-completion bound and the evaluated mandatory-prefix price bounds.

**Proposition 7.2** (Certified elimination of irrelevant candidates). *Let  $L$  be the value of an original-feasible policy and let  $G \geq G(\mathcal{A})$  be a certified free-catalog upper bound. A candidate  $a_i$  satisfying*

$$G - \rho_i < L \quad (44)$$

*belongs to no optimal book. Deleting any collection of such candidates preserves the optimum, the promise, and the symbol budget. Within a search state the corresponding sufficient test is  $G(S) - \sum_{x \in P} \rho(x) - \rho_i < L$  for an undecided candidate.*

*Proof.* A book using  $a_i$  has gross value at most  $G$  and pays at least  $\rho_i$ , since every other charge is nonnegative. Its net value is strictly smaller than an available feasible value  $L$ . The same argument subtracts the mandatory charges in a state. Simultaneous deletions are safe because each excluded book is dominated by the retained feasible value, not by another deleted candidate.  $\square$

Start with the original catalog or a certified screened catalog. At an unresolved state, decide whether to reject or accept its next undecided level. The children partition its completions. An infeasible anchor eliminates a state; a state whose upper bound is no greater than the incumbent is closed; a fully specified feasible book is evaluated exactly. Best-bound processing is useful but not required. A run stopped by a node or time allowance returns the best feasible value and the largest upper bound among its open leaves. Time is checked between atomic oracle calls; measured elapsed times, including any overshoot, are reported.

**Theorem 7.3** (Exact budget-parameterized design and interruption certificate). *For rational quadratic primitives and nonnegative rational charges, the selected-prefix algorithm returns a valid original-space rational interval at every interruption. With no interruption it returns the exact global finite-catalog optimum. Write*

$$Q_m(N) = \sum_{s=0}^{\min(m,N)} \binom{N}{s}. \quad (45)$$

*The unpruned binary cover has at most  $Q_m(N)$  leaves and  $2Q_m(N) - 1$  nodes. With a fixed number  $q_0$  of price calls per node, a sufficient arithmetic bound, including fixed-book allocation, is*

$$O\left((Q_m(N) + mN)[kN + k \log(k+1) + q_0(k+m)N^2]\right). \quad (46)$$

*Consequently the problem is exactly polynomial-time solvable for every fixed  $m$ , with arbitrary  $E$ ; this is an XP bound in  $m$ , not an FPT bound. The free-completion variant  $q_0 = 0$  already establishes this statement. Its rational certificate has  $O(k + N + T)$  entries for  $T$  visited nodes. If  $T_p$  nodes retain priced path witnesses, a sufficient certificate-size bound becomes  $O(k + N + T + T_p m N)$  rational entries, with polynomial bit lengths for a fixed or polynomially bounded  $q_0$ .*

*Proof.* Weak duality proves both upper bounds. A selected prefix is an admissible installed book, and its recovered targets and adjacent lotteries meet every original constraint. Include and exclude children

are disjoint and exhaustive. Induction down the tree therefore covers every admissible book either by an infeasibility proof, a pruned upper bound, or an open upper bound. No book is omitted merely because its eligibility partition differs from the incumbent's. Taking the maximum with the feasible lower value gives the claimed interruption interval. At termination every remaining feasible leaf is a fixed book with a supporting allocation multiplier, so its gap is zero or its value is dominated by the incumbent. The global interval consequently has zero width.

For the size bound, a binary path stops after  $m$  acceptances or after all candidates have been decided. It identifies a unique subset with size at most  $m$ , including the empty infeasible subset. Conversely each such subset identifies one leaf, so a full cover has  $Q_m(N)$  leaves. A finite full binary tree has twice as many nodes minus one. Pruning, screening, and infeasibility removal cannot increase this bound. Each free evaluation uses the fixed-book sweep; each priced evaluation uses Theorem 7.1. The greedy initialization tests at most  $mN$  additional books. This proves (46). In particular, for fixed  $m$ ,  $Q_m(N) = O(N^m)$ ; no accuracy grid or enumeration of class means occurs. The elementary enumeration implication was already proved in antecedent Proposition EC.2.2, reproduced in the companion, using fixed-book separable allocation. The new search supplies budget-respecting completion bounds and checkable pruning, rather than claiming that enumeration itself is novel.

A free-bound node needs only its price and upper endpoint. The checker reconstructs its candidate union and evaluates each scalar support at the eligible knots, the cap, and the clipped quadratic stationary point. A priced node additionally supplies two  $m$ -by- $N$  arrays. The checker verifies anchor, transition, stopping, and peak inequalities, including all mandatory-level restrictions. These are upper potentials, not trusted optimizer outputs. The accepted and rejected index sets follow from the binary cover and need not be stored. The final policy uses at most two lottery atoms per history; the checker evaluates its probabilities, service, expectation caps, pointwise ceilings, opening charges, and root equality directly. These observations prove the entry counts. Rational sums, breakpoint divisions, and finitely many bisections have polynomial encoding length in the rational input and the number of price trials. Finally the checker validates the realized cover; it does not purport to prove an a priori tree-count theorem.  $\square$

This result and Theorem 4.2 give complementary exact regimes. Common catalog eligibility permits variable symbol budget without enumerating books. A fixed symbol budget permits arbitrary catalog eligibility without searching an  $E$ -dimensional mean space. When both vary, the present bounds are not a full complexity classification: no hardness or FPT claim is needed for either positive result. In particular, a hardness assertion for a fixed budget of two or three would conflict with (46) under these rational quadratic primitives. A hybrid implementation should use the one-class oracle when applicable and the completion tree when a small alphabet is the more useful parameter.

## 8 Catalog Refinement and Threshold Robustness

Eligibility complexity is a property of a catalog and its thresholds together. It need not measure the distinctions that a selected book actually uses. The following statements separate changes in the partition, changes in the feasible design problem, and changes in a particular algorithm's work.

**Theorem 8.1** (Catalog-dependent and selected-book partitions). *Let  $E(\mathcal{A}, \tau)$  count distinct eligible catalog prefixes in a feasible catalog, and let  $T$  count distinct primitive thresholds. If  $\mathcal{A} \subseteq \mathcal{A}'$  and  $r_0 = |\mathcal{A}' \setminus \mathcal{A}|$ , then*

$$E(\mathcal{A}, \tau) \leq E(\mathcal{A}', \tau) \leq \min\{T, |\mathcal{A}'|, E(\mathcal{A}, \tau) + r_0\}. \quad (47)$$

*A single insertion splits at most one existing class. For every selected book  $c \subseteq \mathcal{A}$ ,*

$$E(c, \tau) \leq \min\{|c|, E(\mathcal{A}, \tau)\}. \quad (48)$$

*If a threshold perturbation leaves every catalog signature unchanged and maintains  $\tau_j \geq b_j$ , the canonical response, optimal value, and optimal canonical policies are unchanged. In contrast, inserting an unused candidate can increase  $E(\mathcal{A}, \tau)$  without changing the optimal value.*

*Proof.* Two thresholds separated by an old candidate remain separated after insertion, proving refinement of the partition. A new candidate  $x$  divides only the old catalog gap containing  $x$ : histories below  $x$  stay on one side and histories at or above  $x$  stay on the other. Thus at most one class splits per insertion. Identical primitive thresholds cannot be separated. Feasibility gives at least one eligible level for every history, so the possible signature values are  $1, \dots, |\mathcal{A}'|$ . These observations prove (47). Restricting a signature to a subset of candidate levels can only merge classes; a book of  $|c|$  levels has at most  $|c|$  nonempty eligible prefixes. This proves (48).

For unchanged signatures, every fixed book has the same eligible prefix on each history. Formula (2) is therefore unchanged. Its canonical lotteries remain feasible, because drawn terminal levels stay eligible and service cases have total  $t_j \leq b_j \leq \tau_j$ . Taking maxima proves the stability statement.

For a concrete insertion example, take two equally weighted histories with caps  $3/10$  and  $2/5$ , thresholds  $9/20$  and  $11/20$ , promise  $B = 1/4$ , and catalog  $\{0, 1/4, 3/4, 1\}$  with zero charges. Both histories see exactly the first two candidates. Insert  $1/2$  with charge 2. With  $f(x) = 2x - x^2/2$  and nonnegative service costs, gross payoff is at most  $f(1) = 3/2$ , whereas the singleton  $\{1/4\}$  earns  $f(1/4) = 15/32$ . Every book containing the new level earns at most  $-1/2$ . Hence the optimum is unchanged but the catalog partition splits into two classes. Proposition 7.2 removes this candidate with a certificate.  $\square$

For a threshold in the half-open gap  $[a_i, a_{i+1})$ , stability means remaining in that same gap, including its left endpoint and excluding its right endpoint. A convenient sufficient condition away from endpoints is a perturbation smaller than the distance to every catalog level. The value is thus constant on each such signature cell, but a small numerical perturbation can cross a cell boundary. Constancy of the *number* of classes alone does not imply constancy of the value.

**Theorem 8.2** (Robust design and affected-history bound). *Suppose each threshold lies in  $[\tau_j^-, \tau_j^+]$ , with  $\tau_j^- \geq b_j$ . Requiring every realization constraint for every threshold vector in this rectangle is exactly the original model evaluated at  $\tau^-$ . It is therefore solved by either exact regime above when its hypotheses hold. Let*

$$J = \{j : \mathcal{A} \cap [0, \tau_j^-] \neq \mathcal{A} \cap [0, \tau_j^+]\}, \quad M_j = f(1) - f(0) + h_j(b_j).$$

For the same catalog, charges, promise, and symbol budget,

$$0 \leq \mathcal{V}_{m,\mathcal{A}}^{\tau^+}(B; \rho) - \mathcal{V}_{m,\mathcal{A}}^{\tau^-}(B; \rho) \leq \sum_{j \in J} \pi_j M_j. \quad (49)$$

The bound is in affected probability mass, not in the numerical width of the threshold intervals.

*Proof.* The intersection of the pointwise ceiling constraints over the rectangle is  $y_j + C_j \leq \tau_j^-$  on every history. No objective coefficient depends on the threshold, so the robust problem is exactly the lower-ceiling problem. Increasing thresholds enlarges the feasible set, proving the lower inequality.

Take an optimal upper-ceiling book and its targets. Its anchor is no greater than every cap, so the same book and targets admit the lower-ceiling canonical policy. On  $j \notin J$  the two responses coincide. On a changed history the upper response is at most  $f(1)$ , whereas the lower response is at least  $f(0) - h_j(b_j)$ : the lowest selected level followed by the necessary service is feasible, and  $h_j$  is nondecreasing. Thus the loss on that history is at most  $M_j$ . The root equality, branch caps, alphabet size, and all charges are unchanged by this reconstruction. Weighted summation and optimization at the lower ceilings prove (49).  $\square$

A uniform Lipschitz bound in the threshold width would be false without a boundary margin. With one history,  $b = B = 1/4$ , catalog  $\{0, 1/2\}$ , zero charges, budget two,  $f(x) = 2x - x^2/2$ , and  $h(y) = \gamma y^2/2$ , the value is  $-\gamma/32$  just below threshold  $1/2$ . At or above  $1/2$ , an equal-probability lottery on the two levels earns  $7/16$ . The jump  $7/16 + \gamma/32$  does not vanish with the perturbation width. The robust lower-ceiling problem and affected-history bound treat this discontinuity directly rather than pooling across a safety boundary.

These results also delimit certificate reuse. A changed catalog or threshold vector is a changed problem input. An old tree is not automatically a certificate for it. Within a signature cell its canonical terms can be reused after the new input is checked; across cells, the relevant supports and coverage must be rebuilt or independently revalidated. Merging histories according to the winning book compresses that policy's representation, but does not prove upper bounds for competing books with different selected prefixes.

## 8.1 A derived stochastic capacity-reservation model

The same correction arises in a capacity-reservation problem with two explicitly different services. At an observed site state  $j$ , a provider reserves a market-facing capacity  $X_j$  from one certified batch catalog and purchases a non-market backup service  $y_j$  before the batch draw. Both count toward a contractual availability obligation; only market-facing capacity is sold to arriving demand. Backup has a separable provisioning cost and is remunerated through the fixed availability contract, not through the market revenue term. The interface can communicate only the installed batch command, with at most  $m$  possible commands.

Let demand  $D$  be independent of the batch draw, with a common distribution on  $[0, 1]$ , and let  $p$  be revenue per served unit. Then

$$f(x) = p \mathbb{E} \min\{D, x\} = p \int_0^x \mathbb{P}(D \geq u) du. \quad (50)$$

The survival function is nonincreasing, so  $f$  is concave; it is strictly increasing whenever demand has positive

survival below the largest capacity. Uniform demand gives  $f(x) = p(x - x^2/2)$ . Local reservation limits impose  $\mathbb{E}(y_j + X_j) \leq b_j$ , engineering limits impose  $y_j + X_j \leq \tau_j$  on every implemented batch, and a pooled availability contract requires  $\sum_j \pi_j \mathbb{E}(y_j + X_j) = B$ . Certifying a batch level incurs its opening charge once. Maximizing expected market revenue less backup and certification costs gives exactly (3), now derived by integrating a stochastic demand model.

Therefore histories with the same eligible certified batches obey the terminal-first identity, and the backup-cost improvement localizes at the last safe selected batch. More generally the mandatory-prefix tree handles different engineering thresholds without averaging safety limits. This is a mathematical application with specified primitives, not calibrated operational evidence. In particular, a model in which backup also earns the same demand-dependent revenue would have a different objective and is not covered by this identification. The companion gives a direct demand-integral check and a numerical capacity instance; no observed demand or estimated engineering thresholds are invented.

## 9 Computational Study

Table 1 separates the proved regimes before comparing implementations. All inputs below are synthetic rational data, not estimated contracts or operational observations. The protocol fixes generators, seeds, insertion paths, node allowances, price-trial allowances, and envelope resolutions before local execution. Its immutable file and hash identify this experiment; it was not a remotely preregistered experiment. Timing comparisons below use the same recorded execution environment. Earlier timings are not relabeled as matched measurements.

### 9.1 Difficult instances and interruption certificates

We retain the six distinct 12-history, nine-candidate, three-symbol instances underlying the preceding multi-class stress requests. Seventy-two new timed requests cross node allowances of 1, 7, 31, and 127 with 0, 2, and 5 local price trials. Six additional completion runs receive larger allowances. Table 2 compares their exact final intervals with the old group-box gaps. All six completion runs close at zero rational width, using at most 21 visited nodes. Their values agree with the previously available exhaustive references: those values are not newly discovered optima. The new evidence is closure by a compact, independently checked selected-prefix cover rather than unresolved group-mean boxes or full book enumeration.

Figure 1 reports the largest certified absolute gap across these six unchanged inputs at each node allowance. It retains every requested allowance, including unsuccessful early interruptions. A positive-width interval is not an exact solution, even when its width meets a requested tolerance. The global endpoints, rather than rounded display values or numerical statuses, determine all exactness labels.

### 9.2 Nontrivial common eligibility and matched class counts

The 20 strict-prefix instances in Table 3 and the companion test common eligibility without making every candidate eligible. The catalog is nonuniform. All ceilings are distinct points strictly inside one interior

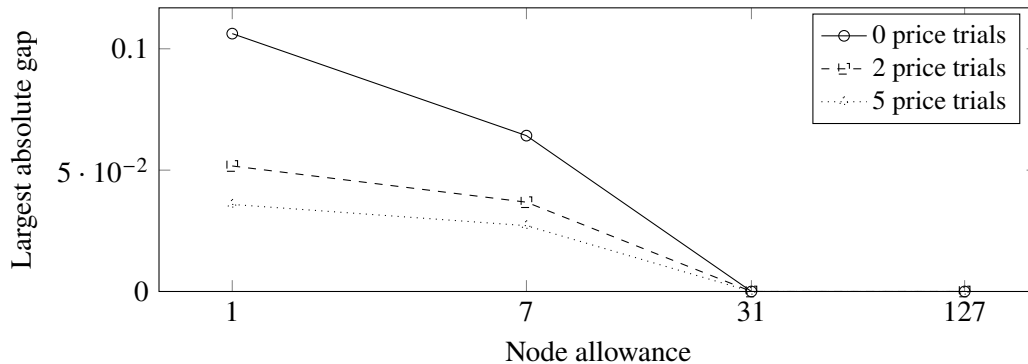


Figure 1: Node-budget and price sensitivity on the six fixed historical stress inputs. The vertical axis is the maximum rational interval width, converted to a display decimal only after verification. All requested budgets, including positive-width interruptions, contribute to the plot.

catalog gap; an entire candidate suffix is ineligible. Caps, probabilities, curvatures, and zero-cost branches remain individual. Promise fractions one third and nine tenths test interior allocations, and the additional 33-level cases vary budgets two and four. Every case is solved exactly by the inherited one-class pooled oracle, including 1,024-history cases. The former all-ceilings-equal-one family remains available as an arithmetic scaling experiment, not evidence of threshold robustness.

Table 4 holds the 16-history, 17-candidate, three-symbol primitives fixed while changing only the ceilings to obtain one through eight classes. Forty-eight requests cross those class counts with three node allowances and two price settings. All 127-node requests close exactly. Local prices reduce nodes in each paired row, but their additional arithmetic makes every displayed priced run slower. Thus fewer nodes are not evidence of lower wall time, and neither these small-class data nor the common-prefix scale experiment establish a universal runtime ranking.

### 9.3 Refinement, screening, and safety boundaries

Table 5 follows six stages of candidate insertion with two opening-charge schedules, each with screening enabled and disabled. In the high-charge path, insertions raise the raw class count from one to four without changing the optimum. Exact screening removes every inserted candidate, retains one active class, and closes at the root throughout. Without screening the search still certifies the same optimum, but visits more nodes. In the low-charge path, new candidates can improve the optimum. This comparison separates harmless class splitting from genuine expansion of the feasible design choices; it does not assume that every newly introduced candidate is irrelevant.

Twelve additional boundary requests place selected subsets of eight thresholds just below or just above a candidate. Three perturbation magnitudes range from  $10^{-2}$  to  $10^{-6}$ . Within a fixed signature cell their exact values agree, while a changed eligible set can change the value. The one-history discontinuity in Section 8 is checked separately. These experiments test the stability statements and do not replace a missing calibrated safety model.

## 9.4 Direct optimization and certificate economics

Table 6 reports the direct mixed-integer formulation on the six difficult inputs and three matched-class inputs. Each tangent envelope and each timed tree receives a separate three-second allowance. Each secant envelope receives a further separate three-second allowance: the two-envelope bracket does not share a single three-second budget. All 18 envelope solves return successful status in this execution. The companion reports both envelopes, wall times including formulation work, primal and dual objectives, residuals, node counts, numerical gaps, and approximation allowances. Exhaustive fixed-book allocation is also rerun on these nine inputs as an independent exact reference. The numerical mixed-integer bounds do not become rational certificates merely because their solver status is successful. Atomic node work and solver preprocessing also mean that nominal wall allowances are not identical stopping mechanisms.

Across the full frozen population, all 182 saved certificates pass independent verification. There are 134 zero-width intervals and 48 open intervals; every open run remains in the companion and raw results. Table 7 reports compressed size, rational bit lengths, and verification-to-optimization time. The largest certificate has 54454 compressed bytes, and the largest numerator and denominator have 96 and 103 bits. Verification is not assumed cheaper than optimization, especially for the one-class exact method. Entry counts and uncompressed sizes are also retained.

The new regression suite checks 664 fixed-book values, 720 mandatory-prefix price values against independent exhaustive calculations, 80 global optima, 40 interruption intervals, and 20 dyadic counts; 15 deliberate corruptions are rejected. Boundary tests add 18 joint comparisons and 276 mandatory-prefix comparisons, including zero caps, zero curvatures, a positive anchor, linear rewards, and budgets through the catalog size. We independently revalidate all 64 inherited pooling certificates and rerun the preceding pooling regression suite. These checks test implemented identities and realized covers; they do not replace the proofs of the complexity counts. The capacity-reservation example additionally verifies its payoff by direct integration of demand, separately from the canonical response formula.

## 10 Conclusion

Exact eligibility pooling converts heterogeneous service allocation into a correction that an ordered command path pays once. It gives an exact finite-catalog joint algorithm when eligibility is common, even with a variable symbol budget. The selected-prefix method complements that result: it handles arbitrary catalog eligibility in polynomial time at every fixed symbol budget, and its price and free-completion bounds certify the original problem at every interruption. The two regimes should be selected according to the structure actually present, not presented as a universal computational ranking.

Catalog refinement can increase a sufficient search dimension without changing the winning policy. Our refinement theorem, threshold-cell analysis, affected-history guarantee, and certified elimination rule make this dependence explicit. The experiments show both nontrivial strict-prefix scaling and exact closure of the inherited small hard instances, while retaining all interrupted runs and their measured gaps. These are reproducible computational observations, not a hardness classification or an application calibration.

An installed alphabet measures the commands available at the terminal interface. It does not count target

tables, encoder logic, read-only constants, probability precision, or certificate storage. Those resources remain separate, and verification costs are reported rather than treated as free. The preserved mathematical archive contains the broader institutional, continuous-design, robustness, and resource-augmentation results under their original assumptions; the current article develops a coherent exact joint-design argument without deleting that record.

**Code and data availability.** The complete code, synthetic inputs, immutable protocol, all successful and interrupted requests, independent certificate checkers, and reproduction instructions are provided in the accompanying code-and-data package under `revisions/or-r51-selected-book-frontier-20260925/`. The root README identifies the current article, companion, and preserved antecedent readers.

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Table 1: Complementary proved regimes for the same finite-catalog problem

| Regime              | Budget   | Guarantee                | Reference     |
|---------------------|----------|--------------------------|---------------|
| Saturated promise   | Variable | Exact polynomial         | Corollary 3.2 |
| Common eligibility  | Variable | Exact polynomial         | Theorem 4.2   |
| Fixed symbol budget | Fixed    | Exact; arbitrary classes | Theorem 7.3   |
| Fixed class count   | Variable | Additive accuracy        | Theorem 6.2   |

The fixed-budget enumeration implication is inherited; the selected-prefix recurrence supplies new bounds and pruning certificates. Fixed-budget polynomial time is an XP statement, not an FPT or variable-budget polynomial-time assertion.

Table 2: Exact closure of the six unchanged historical difficult instances

| Seed  | $E$ | Exact value | Old gap   | Old % | Nodes | Solve (s) | Check (ms) | Bytes |
|-------|-----|-------------|-----------|-------|-------|-----------|------------|-------|
| 49500 | 6   | 29/96       | 0.001771  | 0.59  | 11    | 0.053     | 21.09      | 909   |
| 49501 | 7   | 17569/32800 | 0.001286  | 0.24  | 21    | 0.096     | 32.12      | 1145  |
| 49502 | 6   | 7347/16000  | 0.0009966 | 0.22  | 1     | 0.013     | 4.73       | 716   |
| 49503 | 8   | 199/6600    | 0.001509  | 5.01  | 7     | 0.031     | 10.12      | 774   |
| 49504 | 5   | 19313/41600 | 0.0009165 | 0.20  | 1     | 0.012     | 3.83       | 677   |
| 49505 | 6   | 11181/25600 | 0.005906  | 1.35  | 11    | 0.057     | 20.53      | 914   |

Old gaps are the retained 127-node,  $10^{-3}$  group-box runs, not rerun timings. Their percentages use the known exact value as denominator. New widths are exactly zero in all six cases. The exact values agree with the previously available exhaustive references; the new evidence is certified completion without full enumeration. Bytes are deterministic gzip sizes.

Table 3: Strict common-prefix scaling with distinct interior-gap ceilings

| Histories | Catalog | Budget | Cases | Solve range (s) | Check max (s) | Bytes max | Gap |
|-----------|---------|--------|-------|-----------------|---------------|-----------|-----|
| 16        | 9       | 3      | 2     | 0.017–0.021     | 0.020         | 1512      | 0   |
| 16        | 17      | 3      | 2     | 0.041–0.052     | 0.060         | 2249      | 0   |
| 64        | 9       | 3      | 2     | 0.060–0.077     | 0.079         | 3207      | 0   |
| 64        | 17      | 3      | 2     | 0.143–0.186     | 0.218         | 4647      | 0   |
| 256       | 9       | 3      | 2     | 0.234–0.300     | 0.309         | 9768      | 0   |
| 256       | 17      | 3      | 2     | 0.553–0.731     | 0.854         | 14421     | 0   |
| 1024      | 9       | 3      | 2     | 0.967–1.204     | 1.248         | 34761     | 0   |
| 1024      | 17      | 3      | 2     | 2.255–2.894     | 3.424         | 54454     | 0   |

Each pair uses promise fractions  $1/3$  and  $9/10$ . Catalog levels are squared equally spaced points; every ceiling lies strictly inside the same interior catalog gap and the suffix is ineligible. All caps, weights, and service curvatures remain individual. Four additional 33-level cases cover budgets two and four with 16 or 64 histories; they are also exact and appear in the companion. The old all-ceilings-equal-one family is retained only as an arithmetic scaling check.

Table 4: Matched primitives while varying catalog eligibility classes

| $E$ | Free nodes | Price nodes | Free (s) | Price (s) | Free bytes | Price bytes | Gap |
|-----|------------|-------------|----------|-----------|------------|-------------|-----|
| 1   | 11         | 9           | 0.041    | 0.100     | 1019       | 1196        | 0   |
| 2   | 41         | 25          | 0.086    | 0.373     | 1365       | 1730        | 0   |
| 3   | 11         | 9           | 0.041    | 0.109     | 1023       | 1196        | 0   |
| 4   | 45         | 23          | 0.091    | 0.392     | 1478       | 1970        | 0   |
| 5   | 39         | 23          | 0.086    | 0.360     | 1371       | 1841        | 0   |
| 6   | 37         | 5           | 0.079    | 0.154     | 1379       | 1348        | 0   |
| 7   | 43         | 23          | 0.088    | 0.375     | 1481       | 1940        | 0   |
| 8   | 29         | 15          | 0.068    | 0.224     | 1287       | 1639        | 0   |

Every row has 16 histories, 17 candidate levels, budget three, and the same caps, costs, weights, charges, and promise. Only ceilings change. Price uses at most three local price trials per node; free uses none. Both receive a 127-node, three-second allowance. Prices reduce nodes in every row but increase elapsed time in every row; these data do not support a universal pricing speedup.

Table 5: Harmless class splitting and value-improving catalog refinement

| Inserted | Raw $E$ | Active $E$ | Removed | Raw nodes | Screened nodes | High-charge value | Low-charge value |
|----------|---------|------------|---------|-----------|----------------|-------------------|------------------|
| 0        | 1       | 1          | 0       | 1         | 1              | 0.4407723         | 0.4407723        |
| 1        | 2       | 1          | 1       | 7         | 1              | 0.4407723         | 0.4931528        |
| 2        | 2       | 1          | 2       | 9         | 1              | 0.4407723         | 0.4938506        |
| 3        | 2       | 1          | 3       | 9         | 1              | 0.4407723         | 0.4938506        |
| 4        | 3       | 1          | 4       | 11        | 1              | 0.4407723         | 0.4938506        |
| 5        | 4       | 1          | 5       | 13        | 1              | 0.4407723         | 0.4938506        |

Inserted candidates have charge three in the high-charge path and 1/200 in the low-charge path. The high-charge candidates are never selected; exact screening preserves one active class and the unchanged value. Low-charge candidates can improve the value. All remaining inputs are identical; every displayed value is a zero-width rational optimum.

Table 6: Direct uneliminated mixed-integer diagnostics: tangent envelopes

| Case   | Variables/<br>binaries | Nodes | Solve (s) | Policy value | Solver upper | Solver gap | Envelope error |
|--------|------------------------|-------|-----------|--------------|--------------|------------|----------------|
| S49500 | 249/117                | 1     | 0.023     | 0.3020833    | 0.3020833    | 1.8e-16    | 0.00011        |
| S49501 | 249/117                | 1     | 0.016     | 0.5356402    | 0.5356402    | 0          | 0.00013        |
| S49502 | 249/117                | 1     | 0.007     | 0.4591875    | 0.4591875    | 0          | 4.3e-05        |
| S49503 | 249/117                | 1     | 0.021     | 0.03015152   | 0.03015152   | 0          | 5.9e-05        |
| S49504 | 249/117                | 1     | 0.007     | 0.4642548    | 0.4642548    | 1.2e-16    | 2.8e-05        |
| S49505 | 249/117                | 1     | 0.010     | 0.4367578    | 0.4367578    | 0          | 8.8e-05        |
| E1     | 593/289                | 1     | 0.053     | 0.6183621    | 0.6183621    | 0          | 3e-05          |
| E4     | 593/289                | 1     | 0.066     | 0.6091582    | 0.6091582    | 0          | 3e-05          |
| E8     | 593/289                | 1     | 0.042     | 0.6032692    | 0.6032733    | 0          | 3e-05          |

Every tangent solve and every compared timed tree has a separate three-second allowance. Secant solves have a further separate three-second allowance, not free shared time. All 18 envelope solves return status zero. These are floating-point HiGHS bounds and numerical feasible policies; they are not exact rational certificates. Solver gap concerns the envelope model. Full wall times, secant bounds, residuals, and bracket widths appear in the electronic companion and raw records.

Table 7: Measured certificate economics across all declared runs

| Family     | Runs | Exact | Median bytes | Max bytes | Numerator<br>bits | Denominator<br>bits | Check/solve |
|------------|------|-------|--------------|-----------|-------------------|---------------------|-------------|
| stress     | 72   | 50    | 740          | 1252      | 30                | 30                  | 0.34        |
| closure    | 6    | 6     | 842          | 1145      | 27                | 28                  | 0.34        |
| strict     | 20   | 20    | 4663         | 54454     | 96                | 103                 | 1.07        |
| matched    | 48   | 22    | 1090         | 1970      | 42                | 43                  | 0.35        |
| refinement | 24   | 24    | 845          | 1231      | 30                | 31                  | 0.34        |
| boundary   | 12   | 12    | 734          | 774       | 35                | 36                  | 0.44        |

Bytes are gzip sizes; numerator and denominator columns are maximum bit lengths over all rational entries. Check/solve is the median ratio of independent verification time to optimization time. The raw records additionally report uncompressed bytes and rational-entry counts. The 48 nonzero-width cases remain open, not exact. The strict family uses the one-class pooled certificate; the other families use selected-prefix certificates.